Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY **9700/35**

Paper 3 Advanced Practical Skills 1 **October/November 2021**

CONFIDENTIAL INSTRUCTIONS

7585619210

This document gives details of how to prepare for and administer the practical exam.

The information in this document and the identity of any materials supplied by Cambridge International are confidential and must NOT reach candidates either directly or indirectly.

The supervisor must complete the report at the end of this document and return it with the scripts.

INSTRUCTIONS

- If you have any queries regarding these confidential instructions, contact Cambridge International stating the centre number, the syllabus and component number and the nature of the query.
  email info@cambridgeinternational.org
  phone +44 1223 553554

This document has **8** pages.

© UCLES 2021

General information about practical exams

Centres must follow the guidance on science practical exams given in the *Cambridge Handbook*.

Safety

Supervisors must follow national and local regulations relating to safety and first aid.

Only those procedures described in the question paper should be attempted.

Supervisors must inform candidates that materials and apparatus used in the exam should be treated with caution. Suitable eye protection should be used where necessary.

The following hazard codes are used in these confidential instructions, where relevant:

C	corrosive	**MH**	moderate hazard
HH	health hazard	**T**	acutely toxic
F	flammable	**O**	oxidising
N	hazardous to the aquatic environment		

Hazard data sheets relating to substances used in this exam should be available from your chemical supplier.

Before the exam

- The packets containing the question papers must **not** be opened before the exam.
- It is assumed that standard school laboratory facilities, as indicated in the *Guide to Planning Practical Science*, will be available.
- Spare materials and apparatus for the tasks set must be available for candidates, if required.

During the exam

- It must be made clear to candidates at the start of the exam that they may request spare materials and apparatus for the tasks set.
- Where specified, the supervisor **must** perform the experiments and record the results as instructed. This must be done **out of sight** of the candidates, using the same materials and apparatus as the candidates.
- Any assistance provided to candidates must be recorded in the supervisor's report.
- If any materials or apparatus need to be replaced, for example, in the event of breakage or loss, this must be recorded in the supervisor's report.

After the exam

- The supervisor must complete a report for each practical session held and each laboratory used.
- Each packet of scripts returned to Cambridge International must contain the following items:
  - the scripts of the candidates specified on the bar code label provided
  - the supervisor's results relevant to these candidates
  - the supervisor's reports relevant to these candidates
  - seating plans for each practical session, referring to each candidate by candidate number
  - the attendance register.

© UCLES 2021

Specific information for this practical exam

During the exam, the supervisor or other competent biologist (**not** the invigilator) should obtain the results specified on the supervisor's report by following the relevant steps in the question paper. The results should be recorded in the space provided on the supervisor's report.

Organisation of the exam

- Half the candidates should start on Question 1 and the other candidates should start on Question 2.

- For Question 2, each candidate must have uninterrupted use of a microscope for at least 55 minutes.

Materials to be supplied by Cambridge International

- Slide **M1**

  On receipt of the slides, check that they are labelled **M1** and that no slides are broken. The slides should **not** be viewed in advance of the exam. The material on the slides is confidential and must **not** be disclosed to candidates.

  The number of slides supplied by Cambridge International will be equal to half the candidate entry.

Return of slides to Cambridge International

Immediately after the exam, the slides must be:

- returned to Cambridge International in the boxes in which they were received, using the self-adhesive label supplied. The slides must **not** be included in the packet of scripts.

or

- purchased using the order form enclosed with the slides, which should be completed and returned to Cambridge International. The order form must **not** be included in the packet of scripts. Slides and boxes will be charged at the rate of £3.25 per slide plus £1 per box.

If the slides are not returned or purchased by the deadline stated on the order form, the charge will be £3.75 per slide plus £1 per box.

© UCLES 2021

Materials and apparatus for Question 1

Each candidate will need:

materials and apparatus for each candidate	quantity	✓
Plant extract in a beaker, labelled **P**, provided at room temperature (see **Preparation of materials**)	at least 40 cm^3	
[MH] 3.0% (10 vol) hydrogen peroxide solution in a covered beaker, labelled **H**, provided at room temperature (see **Preparation of materials**)	at least 30 cm^3	
[MH] 0.3% (2 vol) hydrogen peroxide solution in a covered beaker, labelled **U**, provided at room temperature (see **Preparation of materials**)	at least 10 cm^3	
Distilled water, in a beaker, labelled **W**, provided at room temperature	at least 100 cm^3	
10 cm^3 syringes	3	
Beakers, capacity 50–100 cm^3	6	
Beaker, capacity 200–300 cm^3	1	
Plastic or rubber tubing, length approximately 20 cm, to give an airtight fit with the nozzle of a 10 cm^3 syringe	1	
Test-tube, large, capacity approximately 40–55 cm^3	1	
Test-tube rack to hold 1 large test-tube	1	
Glass rod	1	
Container with approximately 400 cm^3 tap water, labelled **For washing**	1	
Container, capacity approximately 400 cm^3, labelled **For waste**	1	
Paper towels	12	
Glass marker pen, permanent	1	
Stop-clock or timer, showing seconds	1	
Gloves	2	
Suitable eye protection	1	

© UCLES 2021

Preparation of materials

The mung beans must be soaked for 24 hours before the exam. You must pre-test the plant extract in advance of the examination in case you need to replace the mung beans.

- **P**, Plant extract solution

  Mung beans must be soaked for approximately 24 hours in tap water.

  On the day of the exam the plant extract solution can be prepared using 10 g of soaked mung beans.

  Put approximately 10 g of soaked mung beans into a container with 100 cm^3 of water, liquidise until a paste is formed.

  Strain the mixture through a fine sieve or muslin and retain the liquid plant extract solution. This may contain small pieces of plant material.

  This should produce at least 80 cm^3 of plant extract solution.

  Pre-test for P:

  Put 2 cm^3 of **P** into a test-tube.

  Put 2 cm^3 of **H** into the same test-tube.

  You should observe bubbles appearing within 30 seconds. If the bubbles take longer than 30 seconds, **either** increase the mass of mung beans **or** replace the mung beans.

[MH] - **H**, 3.0% hydrogen peroxide solution

  This is prepared on the day of the exam. Put 50 cm^3 of 6.0% (20 vol) hydrogen peroxide solution into 50 cm^3 of distilled water. Mix well.

[MH] - **U**, 0.3% hydrogen peroxide solution

  This is prepared on the day of the exam. Put 10 cm^3 of 3.0% (10 vol) hydrogen peroxide solution into 90 cm^3 of distilled water. Mix well.

Materials and apparatus for Question 2

Each candidate will need:

materials and apparatus for each candidate	quantity	✓
Microscope with: ● an eyepiece lens, ×10 magnification ● a low-power objective lens, ×10 magnification ● a high-power objective lens, ×40 magnification	1 between 2	
Slide **M1**	1 between 2	

Preparation of materials

- Microscope

  Any lenses which are **not** ×10 or ×40 should be removed or replaced.

  For each candidate:
  - the microscope must be set up on low power
  - the slide must **not** be on the stage of the microscope.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

Supervisor's report

Syllabus and component number	9	7	0	0	/	3	5
Centre number							

Centre name .....

Time of the practical session .....

Laboratory name/number .....

Give details of any difficulties experienced by the centre or by candidates (include the relevant candidate names and candidate numbers).

You must include:
- any difficulties experienced by the centre in the preparation of materials
- any difficulties experienced by candidates, e.g. due to faulty materials or apparatus
- any specific assistance given to candidates.

© UCLES 2021

[Turn over

Temperature of exam room …….°C

Results for Question **1(a)(ii)**

Results for Question **1(a)(iii)**

Declaration

1 Each packet that I am returning to Cambridge International contains all of the following items:
- the scripts of the candidates specified on the bar code label provided
- the supervisor's results relevant to these candidates
- the supervisor's reports relevant to these candidates
- seating plans for each practical session, referring to each candidate by candidate number
- the attendance register.

2 Where the practical exam has taken place in more than one practical session, I have clearly labelled the supervisor's results, supervisor's reports and seating plans with the time and laboratory name/number for each practical session.

3 I have included details of difficulties relating to each practical session experienced by the centre or by candidates.

4 I have reported any other adverse circumstances affecting candidates, e.g. illness, bereavement or temporary injury, directly to Cambridge International on a *special consideration form*.

Signed ..... (supervisor)

Name (in block capitals) .....

© UCLES 2021